# Sanaul Haque

## Divish_Urvish_Shah_Proposal

Assignment 206

### Document Details

Submission ID
trn:oid:::3117:634941406

Submission Date
Oct 6, 2026, 6:53 PM GMT+6

Download Date
Oct 6, 2026, 6:58 PM GMT+6

File Name
Divish_Urvish_Shah_Proposal.pdf

File Size
204.5 KB

8 Pages

2,587 Words

15,121 Characters

# 25% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

## Match Groups

**45** Not Cited or Quoted 25%
Matches with neither in-text citation nor quotation marks

**0** Missing Quotations 0%
Matches that are still very similar to source material

**0** Missing Citation 0%
Matches that have quotation marks, but no in-text citation

**0** Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

## Top Sources

22% Internet sources
23% Publications
20% Submitted works (Student Papers)

## Match Groups

- **45** Not Cited or Quoted 25%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

## Top Sources

22% Internet sources
23% Publications
20% Submitted works (Student Papers)

## Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

| | | |
|---|---|---|
| 1 | Internet | arxiv.org 2% |
| 2 | Internet | tel.archives-ouvertes.fr 2% |
| 3 | Student papers | Rutgers University, New Brunswick on 2021-04-14 2% |
| 4 | Student papers | Rijksuniversiteit Groningen - Tii on 2026-04-16 2% |
| 5 | Internet | www.research-collection.ethz.ch 1% |
| 6 | Internet | oro.open.ac.uk 1% |
| 7 | Internet | www.osti.gov 1% |
| 8 | Student papers | University of Nottingham on 2025-05-28 1% |
| 9 | Student papers | University of Sydney on 2023-12-27 <1% |
| 10 | Publication | Espen Haug. "Sunyaev–Zel'dovich, Fast Radio Burst, and Metal-Absorption Tests o... <1% |

11 Student papers
University of Nottingham on 2025-05-21 <1%

12 Internet
experts.arizona.edu <1%

13 Student papers
Cardiff University on 2022-09-02 <1%

14 Internet
hdl.handle.net <1%

15 Student papers
Universiteit van Amsterdam on 2019-07-20 <1%

16 Internet
lume.ufrgs.br <1%

17 Student papers
University of Durham on 2023-04-18 <1%

18 Internet
www.ictp-saifr.org <1%

19 Student papers
Boston University on 2017-12-03 <1%

20 Internet
nbi.ku.dk <1%

21 Publication
Suyu, S. H., M. W. Auger, S. Hilbert, P. J. Marshall, M. Tewes, T. Treu, C. D. Fassnach... <1%

22 Internet
hal.science <1%

23 Internet
mdpi-res.com <1%

24 Internet
mediatum.ub.tum.de <1%

25 Publication

Anne Green. "Dark matter in astrophysics/cosmology", SciPost Physics Lecture N... <1%

26 Publication

Pierre Fleury, Julien Larena, Jean-Philippe Uzan. "Line-of-sight effects in strong gr... <1%

27 Internet

www.aertc.org <1%

28 Publication

Hadrien Paugnat, Tommaso Treu, Daniel Gilman. "Macromodel-free flux-ratio pre... <1%

29 Internet

www.annualreviews.org <1%

30 Student papers

Swinburne University of Technology on 2025-05-13 <1%

# *AUDITING ELLIPTICITY-SHEAR DEGENERACY IN STRONG GRAVITATIONAL LENS MODELS*

*A forward-inverse investigation using controlled simulations and RXJ1131-1231 astrometry*



| | |
|---|---|
| **Full Name** | **Divish U Shah** |
| **Institution** | Adani International School |
| **Class / Grade / Year** | Grade 12 (IB Diploma Programme) |
| **Country** | India |
| **Major** | Physics |
| **Submission Date** | 5th October 2026 |

## 1. Research Problem and Question

23 Strong lensing occurs when the density of foreground matter produces an appreciable gravitational deflection of light from a background source that produces multiple images, extended arcs or an Einstein ring. Strong lensing provides a direct look not only into the distribution of luminous matter, but also into the distribution of non-luminous matter, as well as a window into the underlying structure of gravitational systems [1]-[5]. This diagnostic capacity is bounded by a fundamental limitation, as lensing inherently is an inverse problem. Numerous configurations of ellipticity for the lenses, location of the source, radial mass profile, environment perturbations, and line-of-sight structure can yield almost identical images [6]-[10]. The fact that a model is presented graphically is not enough to conclude that the model has been shown to be uniquely supported by the data.

### Central research question

**How reliably can the Einstein radius, ellipticity and external shear of RXJ1131-1231 be distinguished using controlled forward simulations, nested inverse models, profiled q-gamma grids and astrometric perturbation refits?**

### Supporting questions

- How does the combination of Einstein radius, projected ellipticity, source position and external shear affect ring and arc morphologies in a numerically verified lens simulation?
- What measurable characteristics, including arc length, curvature, image count, flux ratio, separation and asymmetry, are best suited to measure controlled parameter changes?
- As the fitted model changes from a bare SIS to an SIE with shear and the observed satellite, how does common-source consistency for RXJ1131-1231 improve?
- What lens parameters are reliably fixed with only point-image astrometry, and which are affected by missing extended-image information and degeneracy?
- Is the trade-off between ellipticity and shear quantifiable using profiled q-gamma grids and perturbation refits?

### Hypotheses

- The imposed Einstein radius will be reproduced by the SIS control simulation within numerical resolution.
- Increasing ellipticity will shorten arc length while increasing image separation and asymmetry.
- External shear will improve the fit of the quadruply imaged system where the angular structure is not captured by the main lens alone.
- Einstein radius and position angle will be recovered more reliably than axis ratio and shear strength.
- Profiling q against gamma will produce an elongated low-residual valley and perturbation refits will show a strong positive correlation.

## 2. Existing Literature and Research Gap

Strong lensing is mathematically well understood, and the lens equation, convergence, critical curves, caustics and magnification are described extensively in the literature [3]-[5]. Elliptical lens models capture anisotropic projected mass more realistically [6]-[8], while external shear is generally introduced as a quadrupole term that accounts for neighboring structure [7], [9]. Combined with dynamical modelling, galaxy-scale lens samples have demonstrated that total density profiles can be recovered reliably when selection effects and source-reconstruction issues are addressed [10]-[13].

5 RXJ1131-1231 is particularly well studied. It was identified as a quadruply imaged quasar with an optical Einstein-ring candidate [14], and later modelled through reconstruction of its lensed source galaxy [15]-[19]. Suyu et al. combined imaging, time delays, stellar kinematics and environmental information to produce a detailed analysis [16]. These published results provide a benchmark for comparison with the deliberately simplified astrometric inverse model in this study.

29 The dark-matter relevance of strong lensing depends on the reliability of the macro model. Substructure studies have used flux anomalies, gravitational imaging and population-level inference [23]-[29]. However, fitted external shear is not necessarily a direct measurement of physical shear; it may absorb deficiencies in an overly rigid mass profile [30]. Unmodelled angular complexity and inflexible source descriptions can also generate convincing false-positive substructure signals [37], [38].

### Research gap

The gap addressed here is therefore not the absence of SIS, SIE or shear formalisms, but the need for a unified and reproducible audit linking controlled morphology to observational inference. A model may recover the Einstein scale while leaving angular parameters non-identifiable. Before residual structure is interpreted as evidence for dark-matter substructure, it is necessary to determine what a smooth macro model can explain, which parameters are robust, and which compensate for one another.

### Original contribution

The study is structured as a two-way experiment. The forward stage generates known mass distributions and identifies observable signatures of individual parameters. The inverse stage tests how much of that parameter space can be recovered from published RXJ1131-1231 astrometry. The central technical contribution is to convert the qualitative concern that ellipticity and shear can look alike into a measurable identifiability test using profiled q-gamma grids and repeated coordinate-perturbation refits.

The dark-matter claim remains deliberately bounded. The study does not claim a subhalo detection, a stellar-halo decomposition, or a dark-matter particle constraint. It establishes the macro-model audit that should precede such claims.

## 3. Methodology I: Controlled Forward Modelling

The relation between image position θ and source position β is described by the reduced lens equation $\beta = \theta - \alpha(\theta)$. The circular control model is the singular isothermal sphere (SIS), whose deflection is fixed in magnitude and radial in direction. For a centered source, the SIS produces an Einstein ring at radius $\theta_e$. The singular isothermal ellipsoid (SIE) introduces the axis ratio q and position angle $\varphi_e$ and reduces continuously to the SIS as q approaches unity. External shear is added as a constant quadrupolar contribution with amplitude γ and angle $\varphi_\gamma$.

### Simulation design

The synthetic image plane uses a 160 × 160 grid with a pixel scale of 0.05 arcsec $pixel^{-1}$. The lens is fixed at the origin, and the principal Einstein scale is $\theta_e = 1.1$ arcsec. The background source is a circular Gaussian with amplitude 1 and width $\sigma_s = 0.075$ arcsec, located by default at $\beta = (0.07, 0.02)$ arcsec. The fiducial SIE-type lens uses ellipticity components $e_1 = 0.16$ and $e_2 = 0.05$, corresponding to ellipticity amplitude $\varepsilon = 0.168$, axis-ratio proxy $q = 0.713$ and position angle $\varphi_e = 8.68°$.

### Experimental sequence

| Experiment | Controlled change | Primary output |
|---|---|---|
| Analytic SIS control | $\beta = (0, 0)$; no shear | Recovered Einstein-ring radius |
| SIS-SIE comparison | Fixed source, grid and normalization | Morphology and residual metrics |
| Ellipticity sweep | $e_1 = \{0.00, 0.08, 0.16, 0.24\}$; $e_2 = 0.05$ | Arc length, images, flux, separation |
| Source sweep | $\beta = \{(0, 0), (0.06, 0.02), (0.14, 0.04), (0.26, 0.06)\}$ | Flux-ratio and asymmetry response |
| Shear sweep | $\gamma = \{0, 0.020, 0.045, 0.060, 0.080\}$; $\varphi_\gamma = 13.3$ | Separation of shear and ellipticity effects |
| Noise test | PSF, sky, Poisson and read noise | Robustness across 30 noise seeds |

### Forward outputs

The forward stage measures Einstein radius, arc length, mean curvature radius, number of recovered images, component flux ratio, maximum separation, residual norms and azimuthal asymmetry. These metrics replace purely visual comparison with quantitative diagnostics. Parameter sweeps vary one factor at a time while holding the remaining settings constant.

## 3. Methodology II: Observational Inverse Modelling

The observational target is RXJ1131-1231, a gravitational lens system with four quasar images, a bright lensed host-galaxy ring, a foreground early-type lens galaxy and a nearby satellite [14]-[16]. The adopted redshifts are z_d = 0.295 for the lens and z_s = 0.658 for the source. The inverse analysis uses the published optimized F814W astrometric coordinates reported by Suyu et al. [16]. Coordinates are translated so that the main lens galaxy lies at the origin while preserving the published orientation convention.

**Nested model hierarchy**

| Model | Free parameters | Purpose |
|---|---|---|
| M1: SIS | $\theta_e$ | Circular baseline |
| M2: SIS + shear | $\theta_e$, $\gamma$, $\varphi_\gamma$ | Tidal quadrupole without intrinsic ellipticity |
| M3: SIE | $\theta_e$, q, $\varphi_e$ | Intrinsic projected ellipticity without shear |
| M4: SIE + shear | $\theta_e$, q, $\varphi_e$, $\gamma$, $\varphi_\gamma$ | Separates intrinsic and external angular structure |
| M5: SIE + shear + satellite | M4 parameters; $\theta_{e,s}$ = 0.20 arcsec fixed | Adds the observed local perturber |

For each observed image position $\theta_i$, the model back-projects $\beta_i = \theta_i - \alpha(\theta_i)$. A correct model maps all four images to a compact common source. The scalar diagnostic is the source-plane scatter:

$$RMS\beta = [(1/2N)\ \Sigma_i\ |\beta_i - \bar{\beta}|^2]^{1/2}.$$

This diagnostic is applied consistently across the nested models for comparison. It is not treated as a full image-plane likelihood, Bayes factor or detection significance.

Parameter bounds and identifiability audit

The circularized Einstein radius is constrained to 0.20–3.00 arcsec, the axis ratio q to 0.20–0.999, and the shear magnitude $|\gamma|$ to 0–0.50. Lens and satellite positions remain fixed at their published values, while orientation angles span 180°. To test the ellipticity–shear degeneracy, $\theta_e$ and both orientation angles are re-optimized on a 43 × 43 grid covering q = 0.70–0.98 and $|\gamma|$ = 0.055–0.190 for both M4 and M5, producing 3,698 profiled fits.

Sensitivity and reproducibility

Forward degradation is repeated across 30 independent noise seeds. The numerical stability of the full inverse model is tested through 1,000 refits after adding Gaussian perturbations of $\sigma$ = 0.005 arcsec to each coordinate of every observed image. Pearson's correlation coefficient and a local least-squares trend in the (q, $|\gamma|$) plane characterize the perturbation cloud without being interpreted as universal lensing laws.

## 4. Practicalities, Limitations and Research Value

### Reproducibility and data provenance

All observational coordinates and benchmark comparison parameters are transcribed from the published RXJ1131-1231 analysis [16]. The computational workflow contains the analytic deflection calculations, optimization bounds, parameter sweeps, critical-curve and caustic calculations, coordinate-perturbation analysis and machine-readable exports required to reproduce the proposed comparisons. No independent redistribution or claiming of HST image pixels is involved.

### Limitations

- Four point-image positions provide substantially less information than extended ring pixels and do not replace a calibrated image-plane likelihood.
- The source-plane RMS is a common-source diagnostic, not a Bayesian evidence value or formal model-selection statistic.
- The coordinate-perturbation scale tests numerical stability and is not a substitute for the complete HST astrometric covariance matrix.
- A fixed isothermal radial profile, simplified PSF treatment and fixed satellite strength limit physical interpretation.
- The q-gamma correlation and profile maps apply only to this data set, model family and perturbation experiment.
- No stellar-mass decomposition, subhalo detection or dark-matter particle constraint will be claimed.

### Expected research value

The project is designed to distinguish robust lens-scale recovery from non-identifiable angular structure. Its value lies in demonstrating, on a realistic strong-lens system, how intrinsic ellipticity, external shear and a known perturber can compensate for one another. This creates a reproducible macro-model audit that can be performed before residuals are interpreted as evidence of dark-matter substructure.

The study also provides a transferable workflow: validate the simulator analytically, isolate parameter effects through controlled forward sweeps, fit a hierarchy of increasingly flexible inverse models, stress-test the solution under perturbations, and profile competing parameters directly. The same logic can be applied to other quadruply imaged systems and to richer extended-image analyses.

### Planned outputs

- A verified set of controlled lens-morphology simulations and quantitative observables.
- A five-model comparison for RXJ1131-1231 based on common-source consistency.
- Profiled q-gamma fit landscapes and a 1000-refit numerical-stability audit.
- A clear boundary separating macro-model validation from unsupported dark-matter claims.
- Reproducible code, parameter tables and machine-readable outputs.

The final interpretation will remain proportional to the available information. A recovered Einstein scale may be robust even when ellipticity and shear are not separately identifiable; this distinction is the central outcome the proposed audit is designed to establish.

## References


[1] T. Treu, 'Strong Lensing by Galaxies,' Annual Review of Astronomy and Astrophysics, vol. 48, pp. 87-125, 2010, doi: 10.1146/annurev-astro-081309-130924.

[2] M. Bartelmann, 'Gravitational Lensing,' Classical and Quantum Gravity, vol. 27, no. 23, art. 233001, 2010, doi: 10.1088/0264-9381/27/23/233001.

[3] P. Saha, D. Sluse, J. Wagner, and L. L. R. Williams, 'Essentials of Strong Gravitational Lensing,' Space Science Reviews, vol. 220, art. 12, 2024, doi: 10.1007/s11214-024-01041-w.

[4] A. J. Shajib et al., 'Strong Lensing by Galaxies,' Space Science Reviews, vol. 220, 2024, doi: 10.1007/s11214-024-01105-x.

[5] S. Vegetti et al., 'Strong Gravitational Lensing as a Probe of Dark Matter,' Space Science Reviews, vol. 220, 2024, doi: 10.1007/s11214-024-01087-w.

[6] R. Kormann, P. Schneider, and M. Bartelmann, 'Isothermal Elliptical Gravitational Lens Models,' Astronomy and Astrophysics, vol. 284, pp. 285-299, 1994.

[7] C. R. Keeton, C. S. Kochanek, and U. Seljak, 'Shear and Ellipticity in Gravitational Lenses,' The Astrophysical Journal, vol. 482, pp. 604-620, 1997, doi: 10.1086/304172.

[8] N. Tessore and R. B. Metcalf, 'The Elliptical Power Law Profile Lens,' Astronomy and Astrophysics, vol. 580, art. A79, 2015, doi: 10.1051/0004-6361/201526773.

[9] C. R. Keeton and A. I. Zabludoff, 'The Importance of Lens Galaxy Environments,' The Astrophysical Journal, vol. 612, pp. 660-678, 2004, doi: 10.1086/422311.

[10] P. Schneider and D. Sluse, 'Mass-Sheet Degeneracy, Power-Law Models and External Convergence,' Astronomy and Astrophysics, vol. 559, art. A37, 2013, doi: 10.1051/0004-6361/201321882.

[11] L. V. E. Koopmans et al., 'The Sloan Lens ACS Survey. III,' The Astrophysical Journal, vol. 649, pp. 599-615, 2006, doi: 10.1086/505696.

[12] M. W. Auger et al., 'The Sloan Lens ACS Survey. X,' The Astrophysical Journal, vol. 724, pp. 511-525, 2010, doi: 10.1088/0004-637X/724/1/511.

[13] A. Sonnenfeld et al., 'The SL2S Galaxy-Scale Lens Sample. IV,' The Astrophysical Journal, vol. 777, art. 97, 2013, doi: 10.1088/0004-637X/777/2/97.

[14] D. Sluse et al., 'A Quadruply Imaged Quasar with an Optical Einstein Ring Candidate: 1RXS J113155.4-123155,' Astronomy and Astrophysics, vol. 406, pp. L43-L46, 2003, doi: 10.1051/0004-6361:20030904.

[15] J.-F. Claeskens et al., 'Multi-Wavelength Study of the Gravitational Lens System RXS J1131-1231. II,' Monthly Notices of the Royal Astronomical Society, vol. 367, pp. 879-904, 2006, doi: 10.1111/j.1365-2966.2006.10024.x.

[16] S. H. Suyu et al., 'Two Accurate Time-Delay Distances from Strong Lensing: Implications for Cosmology,' The Astrophysical Journal, vol. 766, art. 70, 2013, doi: 10.1088/0004-637X/766/2/70.

[23] S. Mao and P. Schneider, 'Evidence for Substructure in Lens Galaxies?' Monthly Notices of the Royal Astronomical Society, vol. 295, pp. 587-594, 1998, doi: 10.1046/j.1365-8711.1998.01319.x.

[24] N. Dalal and C. S. Kochanek, 'Direct Detection of Cold Dark Matter Substructure,' The Astrophysical Journal, vol. 572, pp. 25-33, 2002, doi: 10.1086/340303.

[25] S. Vegetti and L. V. E. Koopmans, 'Bayesian Strong Gravitational-Lens Modelling on Adaptive Grids,' Monthly Notices of the Royal Astronomical Society, vol. 392, pp. 945-963, 2009, doi: 10.1111/j.1365-2966.2008.14005.x.

[26] S. Birrer, A. Amara, and A. Refregier, 'Lensing Substructure Quantification in RXJ1131-1231,' Journal of Cosmology and Astroparticle Physics, no. 05, art. 037, 2017, doi: 10.1088/1475-7516/2017/05/037.

[27] J. Brehmer et al., 'Mining for Dark Matter Substructure,' The Astrophysical Journal, vol. 886, art. 49, 2019, doi: 10.3847/1538-4357/ab4c41.

[28] S. Wagner-Carena et al., 'From Images to Dark Matter,' The Astrophysical Journal, vol. 942, art. /5, 2023, doi: 10.3847/1538-4357/aca525.

[29] N. Anau Montel et al., 'Estimating the Warm Dark Matter Mass from Strong Lensing Images,' Monthly Notices of the Royal Astronomical Society, vol. 518, pp. 2746-2760, 2023, doi: 10.1093/mnras/stac3217.

[30] A. Etherington et al., 'Strong Gravitational Lensing's External Shear Is Not Shear,' Monthly Notices of the Royal Astronomical Society, vol. 531, pp. 3684-3700, 2024.

[37] Q. He et al., 'Testing Strong-Lensing Subhalo Detection with a Cosmological Simulation,' Monthly Notices of the Royal Astronomical Society, vol. 518, pp. 220-244, 2023, doi: 10.1093/mnras/stac2779.

[38] N. Ephremidze et al., 'Dark Matter Substructure or Source-Model Systematics?' Monthly Notices of the Royal Astronomical Society, vol. 542, pp. 2610-2638, 2025, doi: 10.1093/mnras/staf1366.